

Session 3:

Internet & Networking Basics
Internet Fundamentals: How computers communicate.
Protocols: Define HTTP, HTTPS, FTP, SMTP (overview only, 5 mins each).
Domain & DNS: Explain how domain names map to IP addresses. Demonstrate using nslookup google.com. Explain DNS hierarchy: root, TLD, domain, subdomain.
IP Addressing: What is an IP (IPv4 vs IPv6)?
HTTP Lifecycle: Request ? Response ? Status Codes (focus on 200, 404, 500).
Trainer Demo: Use browser DevTools ?
Network tab

1) Full Forms and One-Line Uses

Protocol	Full Form	Use
HTTP	HyperText Transfer Protocol	Used to transfer web pages between a browser and a web server.
HTTPS	HyperText Transfer Protocol Secure	Secure version of HTTP that encrypts data using SSL/TLS.
FTP	File Transfer Protocol	Used to upload and download files between computers over a network.
SMTP	Simple Mail Transfer Protocol	Used for sending emails between mail servers and email clients.

2) Difference Between IPv4 and IPv6 (2 Lines)

- IPv4 uses 32-bit addresses (e.g., 192.168.1.1) and supports about 4.3 billion unique addresses.
- IPv6 uses 128-bit addresses (e.g., 2001:db8::1) and provides a much larger address space with improved efficiency and security.

3) nslookup google.com

- Run the following command on your system:

```
</> Bash
```

```
nslookup google.com
```

- The IP address returned may vary depending on your location and Google's load balancing. Example output:

```
google.com
```

```
Address: 142.250.183.14
```

4) HTTP Status Codes

Code	Meaning
200	OK – Request was successful.
301	Moved Permanently – Resource has been moved to a new URL.
403	Forbidden – Server understands the request but denies access.
404	Not Found – Requested page/resource does not exist.
500	Internal Server Error – Server encountered an unexpected problem.

5) DNS Hierarchy Using

www.google.com

|
└─ www → Subdomain
└─ google → Domain Name
└─ com → Top-Level Domain (TLD)
└─ . → Root Domain

Explanation:

- **Root (.):** The highest level in the DNS hierarchy.
- **TLD (com):** Top-Level Domain representing commercial organizations.
- **Domain (google):** Registered name under the TLD.
- **Subdomain (www):** A subdivision of the domain used to identify specific services or hosts.

Hierarchy Diagram:

Root (.)
|
com
|
google
|
www

This hierarchy helps DNS translate human-readable names like **www.google.com** into IP addresses that computers use for communication.